



Mineral Physics Group
Research in Progress Meeting 2026
Wednesday 8th July



12:00-12:50	Registration
12:50-13:00	Introduction
13:00-13:40	Benjamin Hess (Invited) <i>University Of Oxford</i> Thermodynamics of Stressed Crystals: Stress–Composition Coupling and its Geological Consequences
13:40-14:00	Zhuoqi Lucas Li <i>Imperial College London</i> Mineral Boundary Plane Distributions in Basalt and Their Role in Geological Carbon Sequestration
14:00-14:20	Dongwoo Han <i>University Of Oxford</i> The Magnetic Structure of the Serpentinized Mantle Wedge: A Magnetic Fabric Study on the Sanbagawa Belt, Japan
14:20-14:40	Priyanka Pandit <i>Indian Institute Of Science Education And Research (IISER) Kolkata</i> Compositional Control on Hydrogen Incorporation and Deep Volatile Transport in Cold Slabs Across the Akimotoite–Bridgmanite Transition
14:40-15:00	Refreshment Break
15:00-15:40	Holger Stuniz (Invited) <i>University Of Tromsø, University Of Orléans</i> Recent Advances in Experimental Quartz Rheology and Consequences for the Plastic Deformation of Other Silicates
15:40-16:00	Peixue Dou <i>University College London</i> Point Defects as a Control on the Viscosity Jump in the Lower Mantle
16:00-16:20	Estelle Ledoux <i>University Of Oxford</i> The Stishovite to Post-Stishovite Phase Boundary Studied by Stress Cycling Experiments in a Ddac
16:20-16:40	Siddharth Khan <i>Laboratoire Magmas Et Volcans, CNRS</i> Aluminium Drives Hydrogen Incorporation in Pyrolitic Lower Mantle: Constraints from a Combined FTIR-ERDA-Nanosims Analysis
16:40-16:55	Anastasya Shilova <i>Bruker UK Limited</i> Sponsor Presentation
16:55-18:30	Posters and Refreshments

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Mineral Physics Group Research in Progress Meeting 2026 Thursday 9th July



9:00-9:40

Banusha Kugabalan (Invited)
Imperial College London
Determining Palaeostress Signals Using Rock Magnetism.

9:40-10:00

Yi Li
University Of Oxford
Effects of Temperature-Driven Complexion Transitions and Redox Gradients on Grain-Boundary Plane Distributions in Olivine

10:00-10:20

Lei Liu
Institute Of Earthquake Forecasting, Chinese Earthquake Administration
Potential Significance of Iron-Bearing Superhydrous Phase B in the Formation of Mantle Low-Velocity Zones

10:20-10:40

Emily Wackan
University Of Glasgow
Comparisons Between the Trace Element Composition and Magnetic Structure of the Cloudy Zone in the Imilac and Odessa Iron-Nickel Meteorites

10:40-11:00

Brandon Fish
University Of Oxford
Nickel Content of Iron Meteorites

11:00-11:40

Refreshment Break

11:40-12:20

Zhi Li (Invited)
University College London
Predictive Modelling of Chemically Complex Earth-Forming Materials via Deep-Learning-Driven Atomistic Simulations.

12:20-12:40

Billy Rayner
University College London
Liquid Immiscibility in the Fe-O-S Ternary System at High Pressure and Temperature

12:40-13:00

Yuanjing Li
University College London
Light Element Exsolution In Fe-Si-C-(H) System in Planetary Liquid Cores

13:00-14:00

Break

14:00-15:00

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